Topic 12: Entropy and Energetics

12A: Entropy

Students will be assessed on their ability to:

12.1	understand that, since endothermic reactions can occur spontaneously at room temperature, enthalpy changes alone do not control whether reactions occur
12.2	understand entropy as a measure of disorder of a system in terms of the random dispersal of molecules and of energy quanta between molecules
12.3	understand that the entropy of a substance increases with temperature, that entropy increases as solid → liquid → gas and that perfect crystals at zero kelvin have zero entropy
12.4	be able to interpret the natural direction of change as being in the direction of increasing total entropy (positive entropy change), including gases spread spontaneously through a room
12.5	understand why entropy changes occur during: i changes of state ii dissolving of a solid ionic lattice iii reactions in which there is a change in the number of moles from reactants to products
12.6	understand that the total entropy change of any reaction is the sum of the entropy change of the system and the entropy change of the surroundings, summarised by the expression: $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$
12.7	be able to calculate the entropy change of the system for a reaction, ΔS_{system}, given the entropies of the reactants and products
12.8	be able to calculate the entropy change in the surroundings, and hence ΔS_{total}, using the expression $\Delta S_{\text{surroundings}} = \frac{-\Delta H}{T}$
12.9	understand that the feasibility of a reaction depends on: i the balance between ΔS_{system} and $\Delta S_{\text{surroundings}}$, so that even endothermic reactions can occur spontaneously at room temperature ii temperature, as higher temperatures decrease the magnitude of $\Delta S_{\text{surroundings}}$ so its contribution to ΔS_{total} is less *Students should be able to calculate the temperature at which a reaction is feasible*. *Students may also use $\Delta G = \Delta H - \text{T}\Delta S_{system}$ in answers, although this approach is not a requirement of the specification.*
12.10	understand that reactions can occur as long as ΔS_{total} is positive even if one of the other entropy changes is negative

	Further suggested practicals: Investigate chemical reactions in terms of disorder and enthalpy change, including: i dissolving a solid, including adding ammonium nitrate crystals to water ii gas evolution, including reacting ethanoic acid with ammonium carbonate iii exothermic reaction producing a solid, including burning magnesium ribbon in air iv endothermic reaction of two solids, including mixing solid barium hydroxide, $Ba(OH)_2.8H_2O$ with solid ammonium chloride

12B: Lattice energy

Students will be assessed on their ability to:

12.12	be able to define the terms: i standard enthalpy change of atomisation, $\Delta_{at}H$ ii electron affinity iii lattice energy (as the exothermic process for the formation of one mole of an ionic solid from its gaseous ions)
12.13	be able to construct Born-Haber cycles and carry out related calculations
12.14	understand that a comparison of the experimental lattice energy value (from a Born-Haber cycle) with the theoretical value (obtained from electrostatic theory) in a particular compound indicates the degree of covalent bonding
12.15	understand that polarisation of anions by cations leads to some covalency in an ionic bond, based on evidence from the Born-Haber cycle
12.16	be able to define the terms 'enthalpy change of solution, $\Delta_{sol}H$' and 'enthalpy change of hydration, $\Delta_{hyd}H$ of an ion'
12.17	be able to use energy cycles and energy level diagrams to calculate the enthalpy change of solution of an ionic compound, using enthalpy change of hydration and lattice energy
12.18	understand the effect of ionic charge and ionic radius on the values of enthalpy change of hydration and the lattice energy of an ionic compound
12.19	be able to use entropy and enthalpy changes of solution values to predict the solubility of ionic compounds and discuss trends in the solubility of ionic compounds covered in Unit 2
	Further suggested practical Calculate the enthalpy change when a variety of ionic solids are dissolved in water